

Prajwal Khanal

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RESEARCH INTERESTS

I model soil-vegetation-atmosphere interactions, with a focus on water and carbon fluxes. My work spans both ends of the modeling spectrum: process-based land surface models and data-driven deep learning. I am particularly interested in where the two can inform each other.

EDUCATION

Ph.D., Geo-Information Science and Earth Observation (ITC) <i>University of Twente, The Netherlands</i>	Feb 2023 – present
M.Sc., Environmental Engineering <i>Technical University of Munich, Germany. Grade 1.4/5.0 (German scale; 1.0 = best)</i>	Nov 2020 – Dec 2022
B.E., Civil Engineering <i>Institute of Engineering, Tribhuvan University, Nepal. Rank 20/192 (77.7%)</i>	Nov 2014 – Aug 2018

PUBLICATIONS

Peer-reviewed

- Khanal, P.**, et al. (2024). Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning. *Biogeosciences*, 21, 1533–1548. [doi:10.5194/bg-21-1533-2024](https://doi.org/10.5194/bg-21-1533-2024)
- Khanal, P.**, et al. (2020). Flood inundation mapping of the Babai River for different return periods using HEC-RAS and GIS. *Journal of the Institute of Engineering*, 15(2). [doi:10.3126/jie.v15i2.27639](https://doi.org/10.3126/jie.v15i2.27639)

Under review

- Khanal, P.**, et al. Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model. *Submitted*.
- Khanal, P.**, et al. Deep learning soil moisture and fluxes with multimodal foundation-model embeddings. *In review*.

Conference contributions

- Khanal, P.**, et al. (2026). Bridging the scale gap: daily 10-meter soil moisture, evapotranspiration, and carbon fluxes via multimodal deep learning. *Workshop on Machine Learning for Land and Hydrology*, ECMWF, Reading, UK.
- Khanal, P.**, et al. (2023). Near-surface vs. sub-surface soil moisture impacts on vegetation functioning. *EGU General Assembly*, Vienna, Austria, EGU23-7670. [doi:10.5194/egusphere-egu23-7670](https://doi.org/10.5194/egusphere-egu23-7670)
- Khanal, P.** (2019). Flood inundation mapping: an inference on the outbreak of flood-borne diseases. *International Conference on Health Engineering in Disaster*.

WORK EXPERIENCE

Research Intern, Max Planck Institute for Biogeochemistry, Jena, Germany • Quantified the relative control of near-surface soil moisture vs. terrestrial water storage on global vegetation functioning using satellite observation only. (published in <i>Biogeosciences</i>).	Nov 2020 – Feb 2022
Student Research Assistant, Chair of Hydrology and River Basin Management, TUM • Produced flood inundation maps for the Ouémé River, Benin (FURIFLOOD) and the Naryn River, Kyrgyzstan (ÖkoFlussPlan) using HEC-RAS 1D unsteady-flow models.	Nov 2021 – Mar 2022
Working Student, Jena-Geos Ingenieurbüro GmbH, Munich, Germany • Built an Excel/VBA tool for greenhouse-gas accounting across stationary combustion, refrigeration, and transport.	May 2021 – Dec 2021
Civil Engineer (Freelance), Executive Consulting Engineering and Planner (ECEP), Kathmandu, Nepal • Prepared business plans for community-based water supply projects in Nepal, for the <i>Department of Water Supply and Sanitation, Government of Nepal</i> . • Developed a smart water supply master plan for the development of Lumbini as a smart city, for the <i>Department of Urban Development and Building Construction, Ministry of Urban Development, Nepal</i> .	2018 – 2022

TEACHING EXPERIENCE

Teaching Assistant, Technical University of Munich • Hydrological and Environmental River Basin Modeling; Integrated Water Resources Management; Rapidly Varying Flow (Hydraulics).	2021 – 2022
Assistant Lecturer, Cosmos College of Management and Technology, Lalitpur, Nepal • Designed and delivered an elective course on Python and GIS for water resources (final-year students); taught tutorials and labs for Fluid Mechanics and Hydraulics.	Nov 2018 – Jan 2021

AWARDS & SCHOLARSHIPS

- Second place**, Hackathon on Machine Learning for Earth System Modeling, Aug 2026. Project: *Quantization in foundation models*. Supported by CESOC, ECMWF, TRA Modelling (Univ. Bonn), Forschungszentrum Jülich and Univ. Cologne.
- Deutschland stipendium**, national merit scholarship for M.Sc. studies, TU Munich, 2021/22.
- IOE Merit-Based Scholarship**, merit scholarship for B.E. studies, Tribhuvan University, 2014–2018.

TECHNICAL SKILLS & LANGUAGES

Programming	Python, R, MATLAB, VBA, Bash
ML & data	PyTorch, scikit-learn
Geospatial	Google Earth Engine, GIS, GDAL, HEC-RAS
Tools	Git, HPC/SLURM, LaTeX
Languages	Nepali (native), English (C2), Hindi (fluent), Dutch (A2), German (A1)